

FINE 3300 (Fall 2025): Assignment #2

Important

- Assignment is due on November 10, 2025 by 8:30 am (Toronto – Eastern Standard Time)
- Upload your Python files (.py), output in Excel (.xlsx), figures (.png) and any write-up (.pdf) on Canvas.
- You may make multiple submissions online, but only the most recent submission will be graded.
- Late submissions will automatically receive a grade of *zero*.
- Your code should be commented appropriately. Specify the question number in the comments followed by your code that generates the results.
- Cite any material or source you have referenced.

Introduction

For this assignment you will work with files, NumPy , Matplotlib and Pandas libraries. You will transform the data and extract key pieces of information.

Part A: Loan Amortization and Payment Schedule

One of the documents you receive as part of your mortgage documents from the bank is a loan amortization and payment schedule. For this part of the assignment, expand the class you created in Assignment 1 to generate a loan payment schedule.

Modify your code to also prompt the user for the term of the mortgage. Create a method (i.e., a function within your class) that uses a data frame to build out a payment schedule. Your payment schedule should include the period, starting balance, interest amount, payment, and ending balance. Recall that you have six payment options and must create six data frames. Save each data frame into a single Excel file (with multiple worksheets labelled appropriately). Using Matplotlib, generate a single graph depicting loan balance decline (all 6 plots with their legends). Use the `savefig(...)` function within Matplotlib to save the graph as a PNG file.

Submit your code file, Excel file, and PNG file on Canvas.

Part B: Consumer Price Index

The Consumer Price Index (CPI) represents the price level of a basket of goods and services experienced by the household. In Canada, Statistics Canada tracks the changes to CPI on a monthly basis. We can use the changes to the CPI levels to estimate inflation. Additional information about the CPI and its constituents is available at https://www.statcan.gc.ca/en/subjects-start/prices_and_price_indexes/consumer_price_indexes.

For this assignment, you are provided with 11 data files, consisting of monthly CPI levels for 2024 for 10 provinces and Canada overall. These files have CPI data obtained from Statistics Canada (data obtained from <https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=1810000401>). Read the 11 files into Python and form a data frame.

Python Implementation

1. Combine the 11 data frames into one data frame with column headings: **Item, Month, Jurisdiction, CPI**. For example, if you print the first few entries within the new data frame, they should appear as:

Item	Month	Jurisdiction	CPI
All-items	Jan-24	Canada	158.3
Food	Jan-24	Canada	188.1
Shelter	Jan-24	Canada	178.4

2. Print the first 12 lines of your new data frame
3. For Canada and each of the provinces, report the average month-to-month change in food, shelter, All-items excluding food and energy. Report your numbers as a percent up to one decimal place.
4. Which province experienced the highest average change in the above categories?
5. Using the All-items CPI number, what would be the equivalent salary to \$100,000 received in Ontario in all other provinces (as at December 2024)?
6. Read the `MinimumWages.csv` file. On a nominal basis, which province has the highest minimum wage? Which province has the lowest minimum wage? Using the CPI numbers for December 2024, determine the province with the highest *real* minimum wage?
7. Compute the annual change in CPI for services across Canada and all provinces. Report your numbers as a percent up to one decimal place.
8. Which region experienced the highest inflation in services?
9. In your write-up, answer the above question and paste the output of your Python code.
10. Submit your write-up and coding files on Canvas.